# DASRSNet: Multitask Domain Adaptation for Super-Resolution-Aided Semantic Segmentation of Remote Sensing Images

Yuxiang Cai, Yingchun Yang, Yongheng Shang, *Member, IEEE*, Zhengwei Shen, and Jianwei Yin

***Abstract*— Unsupervised domain adaptation (UDA) has become an important technique for cross-domain semantic segmentation (SS) in the remote sensing community and obtained remarkable results. However, when transferring from high-resolution (HR) remote sensing images to low-resolution (LR) images, the existing UDA frameworks always fail to segment the LR target images, especially for small objects (e.g., cars), due to the severe spatial resolution shift problem. In this article, to improve the segmentation ability of UDA models for LR target images and small objects, we propose a novel multitask domain adaptation network (DASRSNet) for SS of remote sensing images with the aid of super-resolution (SR). The proposed DASRSNet contains domain adaptation for SS (DASS) branch, domain adaptation for SR (DASR) branch, and feature affinity (FA) module. Specifically, the DASS and DASR branches share the same encoder to extract the domain-invariant features for the target and source domains, and these two branches utilize different decoders and discriminators to conduct cross-domain SS task and SR task, which align the domain shift in output space and image space, respectively. Finally, the FA module, which involves the proposed FA loss, is applied to enhance the affinity of SS features and SR features for both source and target domains. The experimental results on the cross-city aerial datasets demonstrate the effectiveness and superiority of our DASRSNet against the recent UDA models.**

***Index Terms*— Adversarial learning, multitask learning (MTL), remote sensing images, semantic segmentation (SS), super-resolution (SR), unsupervised domain adaptation (UDA).**

## I. Introduction

With the continuous advancement of deep learning technologies, convolutional neural networks (CNN)-based [1], [2], [3] semantic segmentation (SS) has become a significant image interpretation approach and widely applied to practical uses in the remote sensing community, e.g., urban management, climate change assessment, and agriculture monitoring. Although these CNN-based models have obtained remarkable results in remote sensing when the models are trained and tested on the same dataset [4], their segmentation performance will degrade significantly when transferring these models from the training dataset (source domain) to classify the unlabeled remote sensing images acquired from various sensors or cities (target domain). This is caused by the distribution discrepancy (domain shift) between the source and target domains [5], [6], [7], [8]. In the field of remote sensing, the domain shift between different domains usually results from numerous factors [9] while capturing remote sensing images, such as imaging time, season, angle and illumination, images acquisition location, and sensor. These factors will significantly affect the distribution discrepancy and latent features of remote sensing images.

Recently, unsupervised domain adaptation (UDA) has been proposed to deal with the aforesaid domain shift instead of manual labeling for target images and has reached impressive performances for cross-domain SS. Generally speaking, most of the UDA frameworks learn to align the target domain and source domain in the image, feature, or output space. For example, Benjdira et al. [10] trained a bidirectional image translation network (e.g., CycleGAN [11]) to alleviate the domain shift between the target domain and the source domain in the image level and then trained the target classifier on the target-like domain, which has a similar data distribution to the real target domain and has the same semantic contents (labels) as the source domain. Similarly, with the help of the image translation model, Zhao et al. [12] aligned the target and source domains not only in the image space but also in the feature space, which better eliminate the domain shift between multiple domains. Without utilizing image translation models, Zhang et al. [13] proposed to adopt a discriminator and a classifier to align the feature space via adversarial learning. Specifically, the discriminator was trained to judge whether the extracted features are from the source domain or the target domain, while the classifier learned the domain-invariant features so as to cheat the discriminator. On top of that, Tsai et al. [14] aligned the source and target domains in the output space of classifier rather than the feature space, which reaches comparable performance. In consideration of all

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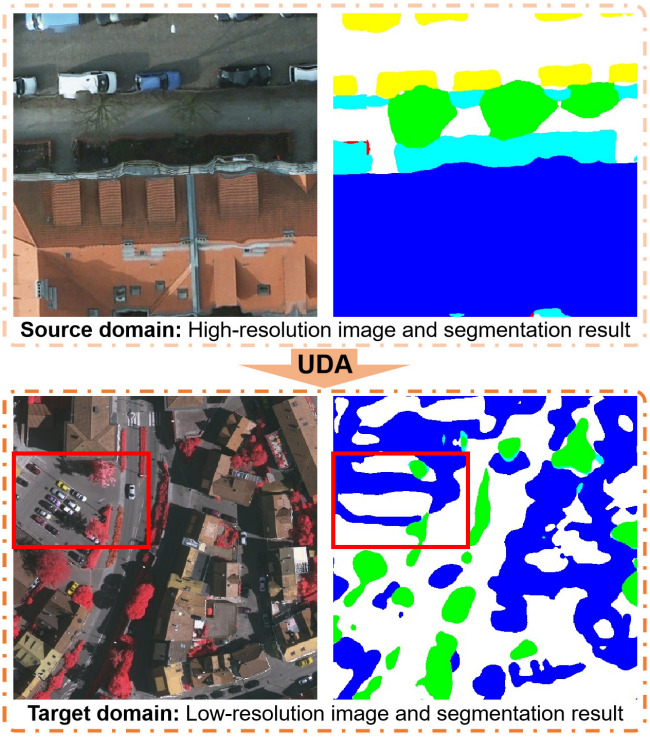


Fig. 1. Sample of spatial resolution shift problem and severe domain shift for small objects. When transferring segmentation model from the source HR remote sensing images (top) to target LR images (bottom) with a large scaling factor of 4, recent UDA methods usually fail to solve the domain shift resulted from huge difference in spatial resolution, especially for small objects (e.g., cars) in remote sensing images as indicated by the red rectangle.

the feature space, output space, and image space, Peng et al. [6] fully matched the target domain with the source domain in all the spaces, which could solve more difficult domain shift issues.

Although the recent UDA models have reached remarkable performances for cross-domain SS tasks, the vast majority of UDA methods do not explicitly consider the spatial resolution shift problem between the target domain and the source domain. Specifically, some UDA methods directly neglect the spatial resolution shift between source and target domains when their differences in spatial resolution are not serious [5], [6], [15], [16], [17], while others handle the spatial resolution shift by scaling the target images and source images to the same spatial resolution via simply interpolation [18], [19], [20], [21]. However, downsampling the HR images to match their spatial resolution with the low-resolution (LR) images will lose the HR textures and features, which are of great importance to SS tasks. Therefore, when transferring from high-resolution (HR) source domain to LR target domain, the majority of recent UDA models always fail to segment the LR target images in remote sensing, especially for small objects within remote sensing images. For example, as shown in Fig. 1, the UDA model outputs terrible predictions for the LR target image and completely omits small objects (e.g., cars), which mainly results from the spatial resolution shift problem between different domains and the serious domain shift for small objects.

In the remote sensing community, the spatial resolution shift problem is usually caused by the fact that the remote sensing images acquired from various sensors generally have various spatial resolutions. For instance, some aerial images have a spatial resolution of 5 cm, while the spatial resolution of other aerial images is 20 cm. In this case, the same object, such as a car, will have different sizes (number of pixels) in two remote sensing images with various spatial resolutions, and the remote sensing images with lower spatial resolutions will be more difficult to segment because of weaker features and smaller objects. These factors will significantly impact the segmentation performance when performing UDA between remote sensing images with various spatial resolutions. Therefore, how to solve the spatial resolution shift problem is an important issue, which needs to be taken seriously in real applications in the remote sensing community.

In field of supervised learning, in order to better extract the features of the small objects within LR remote sensing images, some researchers [22], [23], [24], [25], [26] proposed to combine SS with super-resolution (SR) [27], [28], [29], which aims at recovering HR images from the corresponding LR images, because these two tasks can promote each other [30], [31], [32], [33]. For instance, Xu et al. [22] proposed to integrate the SS network and SR network sequentially into a unified framework so as to better classify LR remote sensing images. Wang et al. [23] proposed a two-stream SS and SR method: one branch is used to segment the LR images, while the other branch is applied to recover the HR images. Moreover, these two branches share the same encoder to extract SS features and SR features. Although these SR-aided SS models have reached apparent performance improvements for segment LR images, they cannot be directly adopted to the UDA tasks because they are trained in a supervised manner.

In this article, to solve the spatial resolution shift problem and improve the UDA ability for LR target images and their small objects, we propose a novel multitask domain adaptation framework named DASRSNet for SR-aided SS of remote sensing images. The proposed DASRSNet has three key parts: domain adaptation for SS (DASS) branch, domain adaptation for SR (DASR) branch, and feature affinity (FA) module. Specifically, the DASS branch trains the shared encoder and segmentation decoder to conduct SS and align the target domain with source domain in the output space via adversarial learning. At the same time, the DASR branch trains the shared encoder and SR decoder to recover HR target images and align different domains in the image space by adversarial learning. We note that the DASS branch and DASR branch share the same encoder to extract the domain-invariant SS and SR features for the target and source domains. Finally, we apply the FA module, which includes the proposed FA loss, to improve the correlation between SS features and SR features for both source and target images.

In summary, our contributions can be concluded as follows.

1) We propose a novel multitask UDA network called DASRSNet for cross-domain SS (SS) of LR remotely sensed images. Our DASRSNet includes three modules, with DASS branch to perform SS and UDA in the output level, DASR branch to conduct SR and UDA in the

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image space, and FA module to boost the correlation between DASS and DASR branches. To the best of our knowledge, DASRSNet is the first work on multitask DASR-aided SS of LR remote sensing images.

2) We propose to integrate SR into the popular UDA method, which enables the existing UDA model to better segment LR target remote sensing images and small objects. Extensive experimental results validate the effectiveness and robustness of this SR-aided architecture.
3) We propose a novel FA loss for the SR-aided SS tasks of cross-domain remotely sensed images. The proposed FA loss is designed to strengthen the affinity of SS features and SR features on both source and target domains.

## II. Related Work

### A. UDA Segmentation

The UDA segmentation is proposed to transfer the segmentation model from the labeled source domain to the unlabeled target domain. In this section, we mainly review the generative adversarial network (GAN)-based [34] UDA methods related to our DASS branch, which align source domain and target domain in the output or feature space through adversarial learning [5], [13], [35], [36]. For instance, Zhang et al. [13] trained one discriminator to distinguish the features of source domain from that of target domain, while the segmentation model was encouraged to deceive the discriminator and extract domain-invariant features. Similarly, Tsai et al. [14] aligned the source and target domains in the output space of the classifier rather than feature space, which achieves comparable performance. Based on this fact, Vu et al. [35] proposed to adapt the entropy of the predicted results instead of the predicted probability of the classifier, which obtains further performance improvement. In the field of remote sensing, Yan et al. [5] proposed to apply a discriminator to determine whether two predicted maps belong to the same domain or different domains, which makes better use of both target and source domains. Moreover, Deng et al. [53] and Al Rahhal et al. [37] proposed to align the satellite images acquired from different cities and aerial images obtained from different locations in an adversarial manner, respectively.

Although these exiting UDA approaches effectively mitigate the domain shift between source and target domains, they do not explicitly study and solve the spatial resolution shift problem, especially for small objects.

### B. Super-Resolution

SR [27], [28], [38], which aims to reconstruct HR images from corresponding LR versions, has been widely adopted in natural image reconstruction [27], [28] and remote sensing image rebuilding [39], [40]. In this section, we focus on the CNN-based SR methods related to our DASR branch. As the pioneering CNN-based SR model, Dong et al. [27] utilized a three-layer CNN as the mapping between LR images and HR images, which leads to apparent improvements against the non-CNN approaches. In the remote sensing community, SR has also been applied to restore HR remote sensing images from LR ones and obtained impressive performances. For example, Haut et al. [39] proposed to learn the relationships between LR and HR remote sensing images without using HR images as training samples. Furthermore, some researchers [41], [42] find that when adopting deeper or more complex CNN structures, the SR performances can be further enhanced, but the computational consumption will become higher as a price. To decrease the computational cost, Shi et al. [29] replaced traditional CNN with subpixel CNN and reached more efficient results.

In this article, to balance the SR performance and computational cost, we follow the spirit of simple yet effective approach [29] for the DASR branch.

### C. Multitask Learning

The multitask learning models [22], [23], [24], [25], [26] combining SS and SR tasks have attracted increasing attention since recent studies have shown that SS and SR tasks are mutually beneficial [30], [31], [32], [33]. Pereira and dos Santos [24] united SS and SR models into an end-to-end network so that the SR module can restore HR remote sensing images from the corresponding LR ones, which helps the SS module produce more accurate predictions. Similarly, Lei et al. [25] proposed to integrate SR into the SS model and applied the SR model as the front unit, which boosts the performances of both SS and SR tasks. On this basis, Xu et al. [22] further explored the impact of the location and complexity of SR components on segmentation performance. Moreover, Hao et al. [26] followed the Siamese architecture and proposed an SR-aided classification framework to better segment the LR remote sensing images. Instead of joining SS and SR models in series, Wang et al. [23] proposed to conduct SS and SR via two parallel branches, which share the same encoder to extract the affinity features of SS and SR.

However, all the above models are trained in a supervised manner and cannot be directly utilized for UDA tasks.

## III. Method

In this article, we focus on common UDA and SR problems, where we have HR source images $x_s^{hr} \in \mathbb{R}^{H\times W\times 3}$ with the corresponding $C$-category labels $y_s \in (1, C)^{H\times W\times C}$ and LR target images $x_t^{lr} \in \mathbb{R}^{h\times w\times 3}$ without labels. We downsample HR $x_s^{hr}$ to the same resolution as LR $x_t^{lr}$ and acquire the LR source images $x_s^{lr} \in \mathbb{R}^{h\times w\times 3}$. Our goal is to train a multitask domain adaptation network for cross-domain SS of LR remotely sensed images with the aid of SR.

As shown in Fig. 2, the proposed network includes three components: DASS branch, DASR branch, and FA module. The DASS branch and DASR branch utilize the shared encoder $E$ to extract the affinity features of SS and SR. To align the source and target domains in the output space, the DASS branch trains the shared encoder $E$ and segmentation decoder $S$ via adversarial learning. At the same time, the DASR branch trains the shared encoder $E$ and SR decoder $R$ to align the recovered source and target images in the image space by

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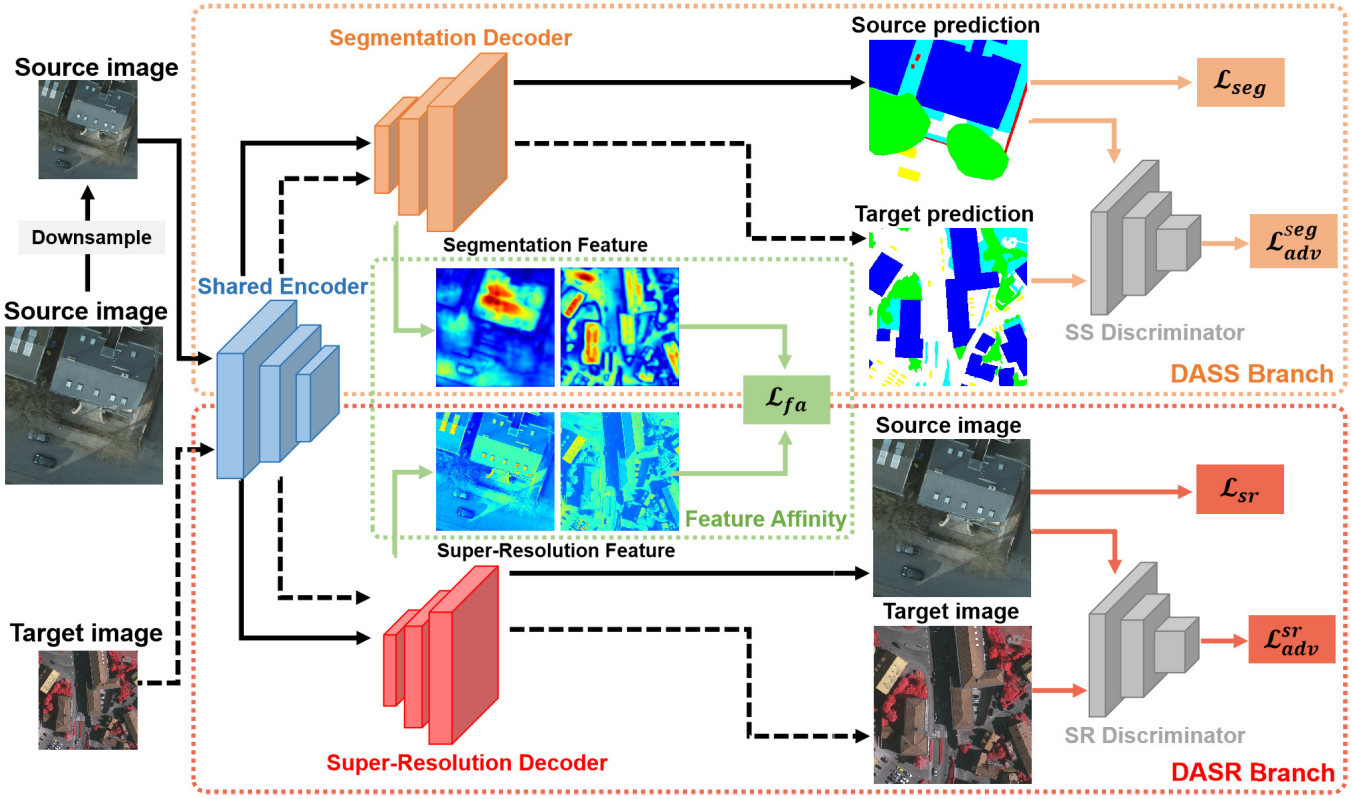


Fig. 2. Overall architecture of the proposed DASRSNet. Our DASRSNet comprises DASS branch, DASR branch, and FA module. The shared encoder $E$ extracts the features for both DASS branch and DASR branch. In the training phase, the proposed DASRSNet takes both source images and target images as input, as illustrated by the black solid arrows (source data flow) and dotted arrows (target data flow). The architecture will be trained by optimizing with three terms: segmentation loss $\mathcal{L}_{\text{seg}}$ and adversarial loss $\mathcal{L}_{\text{adv}}^{\text{seg}}$ for DASS branch, SR loss $\mathcal{L}_{\text{sr}}$ and adversarial loss $\mathcal{L}_{\text{adv}}^{sr}$ for DASR branch, and FA loss $\mathcal{L}_{\text{fa}}$ for FA module. At the test stage, the target images can be sent into the shared encoder $E$ and be segmented and reconstructed by segmentation decoder $S$ and SR decoder $R$ separately, as indicated by the dotted arrows.

adversarial learning. Furthermore, we propose a novel FA module to boost the affinity of SS features and SR features so that these two tasks can better promote each other.

### A. Domain Adaptation for Semantic Segmentation

The DASS branch is designed to eliminate the domain shift between source and target domains for SS task. To make this branch simply but effective, we follow the spirit of the popular UDA approach [35] and align the different domains in the output space by adversarial learning.

First, the shared encoder $E$ and segmentation decoder $S$ form the segmentation network $ES$ to segment both source and target images through learning the domain-invariant features. It is notable that for the segmentation decoder $S$, we append an additional upsampling network [23] containing a few deconvolution layers, BatchNorm and ReLU layers, which do not require many parameters. With the help of upsampling network, the classifier $ES$ takes LR $(h \times w)$ images as input and generates HR $(H \times W)$ predictions for input images. Because of no labels for the target images, we optimize the segmentation network $ES$ (shared encoder $E$ and segmentation decoder $S$) on the labeled source domain through minimizing the class-balanced cross-entropy loss [43]

$$\mathcal{L}_{\text{seg}}(E,S) = -\sum_{h,w}\sum_{c} \alpha_c y_s^{(h,w,c)} \log\left(E(S(x))^{(h,w,c)}\right) \quad (1)$$

where $\alpha_c$ is a weight item for each category $c \in C$. Also, $\alpha_c$ corresponds to the inverse category frequency of the effective number of category $c$ [43], which can be calculated as

$$\alpha_c = \frac{1-\beta}{1-\beta^{n_c}} \quad (2)$$

where $n_c$ is the number of image pixels that belong to category $c$. $\beta = 0.999$ and $\beta$ is a constant. Note that $\beta \to 1$ denotes directly weighting the loss via inverse category frequency of class $c$, while $\beta = 0$ represents no weights, which usually cannot obtain strong performances. Thus, we set $\beta = 0.999$ to achieve a tradeoff between no weights and weighting by an inverse category frequency. Besides, in order to make $\alpha_c$ for different categories in a close range and $\sum_{c=1}^{C} \alpha_c = 1$, we normalize $\alpha_c$ as

$$\alpha_c = \frac{\alpha_c}{\sum_{c=1}^{C} \alpha_c}. \quad (3)$$

As the segmentation network $ES$ of our DASRSNet produces both source predictions and target predictions during the training phase, it is essential to alleviate the domain shift between target and source domains. Hence, we follow the design of the efficient adversarial learning-based UDA models [13], [14], [35] and adopt a discriminator $D_S$ (see Fig. 3) to distinguish the entropy maps of source predicted results of $x_s^{lr}$ from that of target $x_t^{lr}$ while training the segmentation

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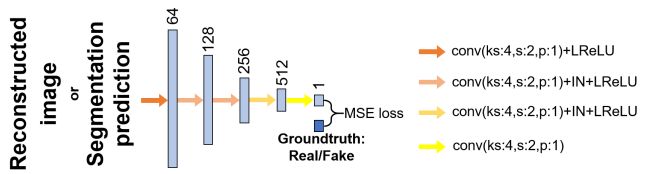


Fig. 3. Structure of discriminators $D_S$ and $D_R$. Conv, $ks$, $s$, and $p$ correspond to convolution layer, kernel size, stride, and padding, respectively. LReLU, MSE, and IN stand for leaky rectified linear unit, mean squared error, and instance normalization, respectively.

network $ES$ to trick $D_S$. The adversarial learning loss $\mathcal{L}_{\text{adv}}^{\text{seg}}$ for adapting the segmentation network $ES$ (shared encoder $E$ and segmentation decoder $S$) and discriminator $D_S$ can be formalized as

$$\mathcal{L}_{\text{adv}}^{\text{seg}}(E, S, D_S) = \sum_{h,w} \log\left(1 - D_S\left(I_s^{(h,w)}\right)\right) + \log\left(D_S\left(I_t^{(h,w)}\right)\right) \quad (4)$$

where $I_s$ corresponds to the entropy maps of $x_s$ and $I_t$ represents the entropy maps of $x_t$. The entropy maps $I$ [35] evaluate the confidence level of the predicted probabilities $E(S(x))$ of the segmentation network $ES$, which takes images $x$ as input. Also, it can be computed as

$$I^{(h,w)} = -\sum_{c} E(S(x))^{(h,w,c)} \log E(S(x))^{(h,w,c)}. \quad (5)$$

Optimizing the segmentation loss $\mathcal{L}_{\text{seg}}$ and adversarial learning loss $\mathcal{L}_{\text{adv}}^{\text{seg}}$ for SS can effectively narrow the domain gap between the source domain and the target domain. However, when transferring from the HR domain to LR domain, there remains a need for more efficient approaches, which can better extract the features of LR target images, especially for small objects.

### B. Domain Adaptation for Super-Resolution

As mentioned before, merely applying the DASS branch is not sufficient to extract and reconstruct HR segmentation features from LR input images, especially for small objects in LR remote sensing images. Since the segmentation decoders always upsample the encoded LR features through simple interpolation methods or uncomplicated networks, which will not produce extra HR textures or features from LR target images. To solve this problem, we take advantage of SR to enhance our DASS branch for the LR target images and small objects. As shown in Fig. 2, we design the DASR branch, which shares the encoder with DASS branch and adopts a simple but effective decoder [29], to reconstruct HR textures and features, which are beneficial to SS tasks, from LR remote sensing images. To provide a better illustration, in Fig. 4, we depict the visual features of DASS branch and DASR branch. From Fig. 4, it can be observed that SR features contain more HR texture information and clearer edge details than SS features. Therefore, the HR SR features reconstructed from the DASR branch can be utilized to lead the DASS branch in learning HR segmentation features [23].

During the training stage, the DASR branch not only learns HR features to help the DASS branch but also aligns the source domain and target domain in the image space by optimizing the shared encoder in an adversarial manner. First, the shared encoder $E$ and the SR decoder $R$ in the DASR branch form the HR image generator $ER$ to extract domain-invariant features in the image level as well as recovering the HR $(H \times W)$ source and target images from LR $(h \times w)$ source and target images. Due to lack of original HR images for the target domain, we optimize the image generator $ER$ (the shared encoder $E$ and the SR decoder $R$) with LR source images $x_s^{lr}$ and corresponding HR images $x_s^{hr}$ according to the mean squared error (mse) loss

$$\mathcal{L}_{\text{sr}}(E, R) = \sum_{h,w} \left\| E\left(R\left(x_s^{lr}\right)\right)^{(h,w)} - \left(x_s^{hr}\right)^{(h,w)} \right\|_2. \quad (6)$$

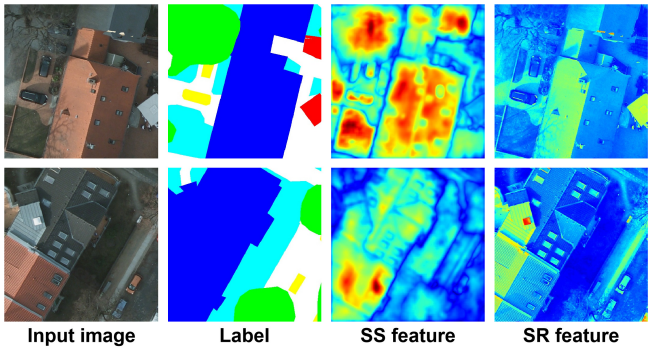


Fig. 4. Visualizations of features of SS and SR tasks for the same input image.

Similar to the DASS branch, we employ a discriminator $D_R$ (see Fig. 3) to align the target domain and source domain in the image space. In particular, the discriminator $D_R$ is trained to differentiate the recovered HR source images $E(R(x_s^{lr}))$ from the recovered HR target images $E(R(x_t^{lr}))$, while the image generator $ER$ learns to cheat $D_R$. The adversarial learning loss $\mathcal{L}_{\text{adv}}^{sr}$ for optimizing generator $ER$ (shared encoder $E$ and SR decoder $R$) and discriminator $D_R$ can be expressed as

$$\mathcal{L}_{\text{adv}}^{sr}(E, R, D_R) = \sum_{h,w} \log\left(1 - D_R\left(E\left(R\left(x_s^{lr}\right)\right)^{(h,w)}\right)\right) + \log\left(D_R\left(E\left(R\left(x_t^{lr}\right)\right)^{(h,w)}\right)\right). \quad (7)$$

Optimizing the SR loss $\mathcal{L}_{\text{sr}}$ and adversarial learning loss $\mathcal{L}_{\text{adv}}^{sr}$ can not only eliminate the domain shift between the source and target domains in the image space but also recover HR features for the DASS branch. In this article, we take SS as the main role and SR as the auxiliary role, and thus, the DASR branch can be dismantled during the test phase of SS task, which will not bring additional computational consumption.

### C. FA Module

As previously discussed, the features extracted from the DASR branch include more fine-grained and complete structural information than features extracted from the DASS branch. For instance, as shown in Fig. 4, the SR features of category cars are clearer and more complete than the SS features, and it is easier to distinguish cars from other

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categories in SR features. Although the more fine-grained and complete structural information in SR features cannot explicitly assign each pixel with a specific category label, this structural information can implicitly cluster the pixels according to the correlations between category and category or object and object. In other words, the similarity between pixels belonging to the same object (e.g., a car) is stronger than the similarity between pixels belonging to different categories. Recent research shows that this structural information could implicitly transfer semantic knowledge, which is conducive to SS tasks [22], [23]. Therefore, though the SS and SR features are with different semantic levels, the learning process of SS tasks can be effectively guided by structural information of SR features.

On this basis, in order to enhance the correlations and relationships between SS features and SR features in multitask UDA training, we propose a novel FA module, which encourages SS features to learn the implicit semantic knowledge from the fine-grained and complete structural information of SR features. Specifically, to model the structural information of features, we utilize a similarity matrix to represent the correlations between inner pixels. Also, for similarity matrix $M$, the pairwise connection $M_{ij}$ between the $j$th and the $i$th pixels of feature map $U$ can be calculated as

$$M_{ij} = \left(\frac{U_i}{\|U_i\|_2}\right)^T \cdot \left(\frac{U_j}{\|U_j\|_2}\right) \tag{8}$$

where $U$ denotes an $H \times W$ SS or SR feature map. We note that there are $HW \times HW$ pairwise connections because we map out the pixel-to-pixel correlations between every two pixels.

Furthermore, by aligning the similarity matrix between DASS and DASR branches, the proposed FA module not only matches SS features and SR features for the same source images but also aligns SS features and SR features for the same target images, as shown in Fig. 2. Such a learning process can be formulated as follows:

$$\mathcal{L}_{\text{fa}}(E, R) = \frac{1}{HW \times HW} \sum_{i=1}^{HW} \sum_{j=1}^{HW} \left\| M_{ij}^{\text{segs}} - M_{ij}^{\text{srs}} \right\|_1 + \left\| M_{ij}^{\text{segt}} - M_{ij}^{\text{srt}} \right\|_1 \tag{9}$$

where $H \times W$ denotes the spatial resolution of SS and SR features. $M^{\text{segs}}$ represents the similarity matrix $M$ of SS feature maps for source images $x_s$, while $M^{\text{srs}}$ is the similarity matrix $M$ of SR feature maps for $x_s$. $M^{\text{segt}}$ corresponds to the similarity matrix $M$ of SS feature maps for target images $x_t$, while $M^{\text{srt}}$ is the similarity matrix $M$ of SR feature maps for $x_t$.

### D. Final Objective Function

With respect to the whole DASRSNet, the final loss function $\mathcal{L}_{\text{DASRSNet}}$ can be concluded as

$$\mathcal{L}_{\text{DASRSNet}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{adv}}^{\text{seg}} \mathcal{L}_{\text{adv}}^{\text{seg}} + \lambda_{\text{sr}} \mathcal{L}_{\text{sr}} + \lambda_{\text{adv}}^{sr} \mathcal{L}_{\text{adv}}^{sr} + \lambda_{\text{fa}} \mathcal{L}_{\text{fa}} \tag{10}$$

TABLE I
PIXEL PERCENTAGES OF DIFFERENT CATEGORIES FOR DATASETS

| Datasets | Class percentages | | | | | |
|---|---|---|---|---|---|---|
| | Imp. surf. | Build. | Low veg. | Tree | Car | Back. |
| Vaihingen | 29.3% | 26.8% | 19.3% | 22.5% | 1.4% | 0.7% |
| Potsdam | 29.8% | 28.1% | 21.0% | 14.5% | 1.8% | 4.8% |

where $\lambda_{\text{adv}}^{\text{seg}}$, $\lambda_{\text{sr}}$, $\lambda_{\text{adv}}^{sr}$, and $\lambda_{\text{fa}}$ represent the hyperparameters for the corresponding losses.

The goal of the final learning objective is to optimize the shared encoder $E$, segmentation decoder $S$, and SR decoder $R$ as follows:

$$E^*, S^*, R^* = \min_{E,S,R} \max_{D_S, D_R} \mathcal{L}_{\text{DASRSNet}}. \tag{11}$$

## IV. EXPERIMENTS AND ANALYSIS

### A. Datasets Description

To verify the effectiveness and robustness of our method, extensive experiments are conducted on open-source benchmark datasets [44], involving aerial images obtained from different cities (Vaihingen and Potsdam). Also, the aerial images are annotated with semantic labels, including backgrounds, low vegetations, trees, buildings, cars, and impervious surfaces. The Vaihingen (VAIH) dataset consists of 33 (near 2000 $\times$ 2000 pixels) images that have three spectral bands (infrared, red, and green bands). The spatial resolution of original (HR) VAIH dataset is 9 cm. To acquire the LR images, we downsample VAIH images via bicubic interpolation [45] with a scaling factor of 2 and obtain the LR VAIH dataset whose spatial resolution is 18 cm. The Potsdam (POTS) dataset includes 36 (6000 $\times$ 6000 pixels) images that have four spectral bands (infrared, red, green, and blue bands). We split the Potsdam dataset into Potsdam IRRG (POTSIRRG) dataset and Potsdam RGB (POTSRGB) dataset according to spectral bands. The POTSIRRG dataset comprises infrared, red, and green bands, while the POTSRGB dataset consists of red, green, and blue bands. Different from the VAIH dataset, the spatial resolution of the original (HR) POTSIRRG and POTSRGB datasets is 5 cm due to differences in imaging sensor. In a similar way, we obtain the LR POTSIRRG and LR POTSRGB datasets through bicubic interpolation [45] with a scaling factor of 4. Thus, their spatial resolution is about 20 cm.

In our experiments, all the images and labels corresponding to the Vaihingen dataset are chipped into 512 $\times$ 512 patches with an overlap of 200 pixels, while all the images and labels corresponding to the Potsdam dataset are sliced into 512 $\times$ 512 patches with no overlap. For each derivative dataset (e.g., HR VAIH dataset), we randomly choose 70% labeled patches as training samples and apply the rest patches as test samples. The category percentages for two basic datasets (Potsdam and Vaihingen datasets) are reported in Table I. It is notable that the category percentages for the six derivative datasets are the same as the two basic datasets. Fig. 5 shows the sampled HR and LR patches for the six derivative datasets.

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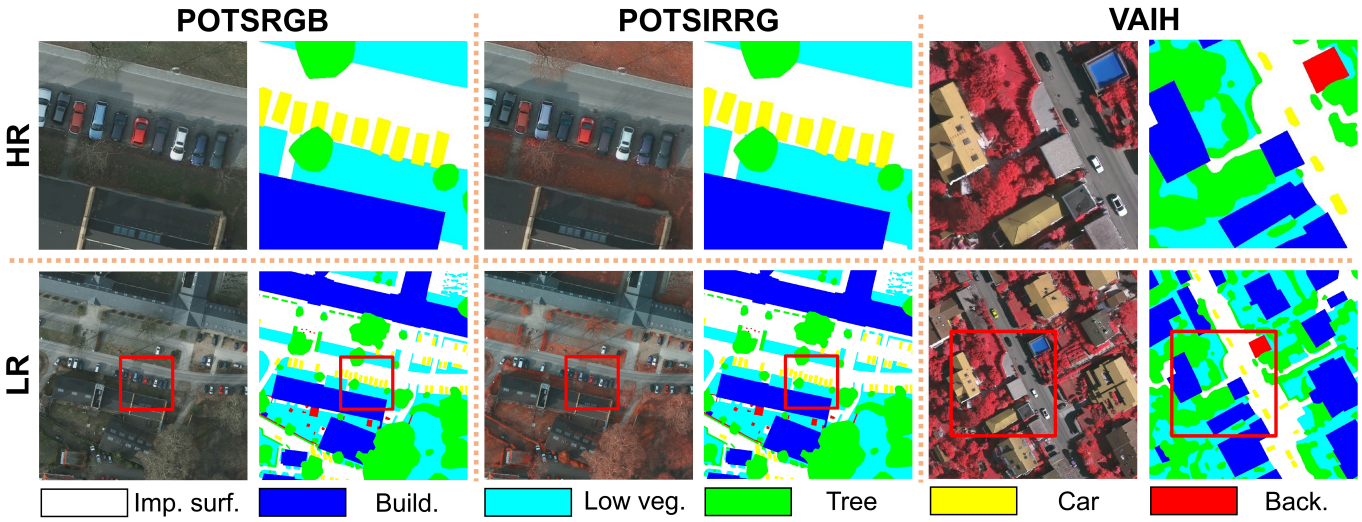


Fig. 5. Patches (512 × 512) and corresponding labels sampled from the HR POTSRGB, LR POTSRGB, HR POTSIRRG, LR POTSIRRG, HR VAIH, and LR VAIH datasets. The top line corresponds to the original HR images and labels. The bottom line denotes the LR images and labels downsampled from the original HR images and labels with a scaling factor of 4 for the POTSRGB and POTSIRRG datasets and a scaling factor of 2 for the VAIH dataset. We note that the HR example (top) and red rectangle part in the corresponding LR example (bottom) depict the same scenario with different resolutions, which will better present the visual difference caused by spatial resolution. Also, we utilize the same scenario for POTSRGB and POTSIRRG datasets to better show the visual difference caused by spectral bands. Blue, white, green, cyan, yellow, and red correspond to buildings (Build.), impervious surfaces (Imp. surf.), tree, low vegetation (Low veg.), car, and background (Back.), respectively.

### *B. Experimental Settings*

*1) Network Architecture:* We adopt the encoder of Deeplabv3+ [3] as the shared encoder $E$, the decoder of Deeplabv3+ [3] as the SS decoder $S$, and the deconvolution architectures of ESPCN [29] as the SR decoder $R$. For the discriminators $D_S$ and $D_R$ of our DASRSNet, we employ the discriminator of BiFDANet [46], which applies five convolutional layers followed by leaky ReLU activation function and instance normalization layer, as shown in Fig. 3.

*2) Training Details:* During the training phase, the scaling factor of DASR branch is set to 2, 2, and 4 for the HR VAIH → LR POTSIRRG experiment, HR VAIH → LR POTSRGB experiment, and HR POTSRGB → LR VAIH experiment, respectively. The encoder $E$, decoder $S$, decoder $R$, discriminator $D_S$, and discriminator $D_R$ are jointly trained for 100 epochs by the Adam optimizer [47] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rates for all the networks are set to $10^{-4}$ utilizing the poly learning rate strategy with power 0.9. Moreover, hyperparameters $\lambda_{\text{adv}}^{\text{seg}}$, $\lambda_{\text{sr}}$, $\lambda_{\text{adv}}^{sr}$, and $\lambda_{\text{fa}}$ are set to 0.01, 0.1, 0.001, and 1, respectively.

*3) Evaluation Metrics:* We adopt F1-score (F1) and intersection over union (IoU) [48] to assess the quantitative SS performance for all the SS approaches and apply peak signal-to-noise ratio (PSNR) [49] and structural similarity index measure (SSIM) [49] to measure the SR performance quantitatively for all the SR methods. The two SS metrics F1 and IoU [48] are calculated as

$$\text{F1} = \frac{2\text{TP}}{2\text{TP} + \text{FN} + \text{FP}} \tag{12}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{13}$$

where FP, TP, and FN represent false positive, true positive, and false negative, respectively.

The two SR metrics PSNR and SSIM are expressed as

$$\text{PSNR}(x, y) = 10 \lg \frac{\left(\text{MAX}_y\right)^2}{\frac{1}{wh}\sum_{i=1}^{w}\sum_{j=1}^{h}\left(x_{ij} - y_{ij}\right)^2} \tag{14}$$

$$\text{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{15}$$

where $y$ denotes the ground truth $w \times h$ image and $x$ is the recovered $w \times h$ image. $\text{MAX}_y$ means the maximum value of $y$. $\mu_y$ and $\mu_x$ correspond to the mean values of $y$ and $x$. $\sigma_y$ and $\sigma_x$ represent the variances of $y$ and $x$, and $\sigma_{xy}$ is the covariance between $y$ and $x$. We apply two constants $c_1$ and $c_2$ to prevent instability.

### *C. Results and Discussion*

In this section, to confirm the superiority and robustness of our proposed DASRSNet, we compare the proposed DASRSNet to numerous recent state-of-the-art (SOTA) UDA approaches: Deeplabv3+ [3], AdaptSegNet [14], Benjdira's [10], AdvEnt [35], TriADA [5], IntraDA [36], BiFDANet [46], IterDANet [16], FDANet [6], SAC [50], CPSL [7], ESPCN [29] + AdvEnt [35], and D-DBPN [42] + AdvEnt [35]. ESPCN + AdvEnt and D-DBPN + AdvEnt denote the combination of SR method [29], [42] and UDA method [35], which first performs SR [29], [42] to recover HR target images and then conducts UDA [35] between source images and reconstructed HR target images. Notably, to make a fair comparison, the training parameters and network architectures of the aforementioned comparison approaches are the same as our DASRSNet.

First, we train our DASRSNet and the comparison models to conduct UDA from the HR VAIH dataset to LR POTSIRRG dataset with small domain shift, which mainly results from the

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TABLE II

SEGMENTATION RESULTS OF THE COMPARING SEGMENTATION MODELS ON THE ADAPTATION FROM THE HR VAIH DATASET TO LR POTSIRRG DATASET. THE SCALING FACTOR BETWEEN SOURCE DOMAIN AND TARGET DOMAIN IS NEAR TWO TIMES. IMP SURF REPRESENTS IMPERVIOUS SURFACES AND LOW VEG REPRESENTS LOW VEGETATION

| Source: HR VAIH, Target: LR POTSRGB | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | Imp surf | | Building | | Low veg | | Tree | | Car | | Background | | Avg | |
| | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | mF1 | mIoU |
| Deeplabv3+(baseline) | 42.3 | 24.3 | 47.2 | 29.9 | 8.4 | 4.3 | 7.9 | 4.1 | 17.8 | 9.9 | 1.6 | 0.8 | 20.9 | 12.2 |
| Benjdira's | 65.1 | 52.8 | 63.7 | 51.9 | 48.1 | 34.1 | 45.8 | 28.7 | 18.3 | 10.1 | 12.9 | 6.9 | 42.3 | 30.8 |
| AdaptSegNet | 68.9 | 51.8 | 70.1 | 53.6 | 57.6 | 39.4 | 34.1 | 20.3 | 27.4 | 17.4 | 9.6 | 4.6 | 44.6 | 31.2 |
| AdvEnt | 70.3 | 53.9 | 71.8 | 56.5 | 53.8 | 36.8 | 40.2 | 26.4 | 25.5 | 16.7 | 10.2 | 5.3 | 45.3 | 32.6 |
| TriADA | 68.6 | 54.7 | 70.5 | 55.2 | **61.8** | **46.8** | **50.1** | **33.8** | 31.3 | 20.8 | 18.7 | 10.3 | 50.2 | 36.9 |
| IntraDA | 68.2 | 54.1 | 70.9 | 56.6 | 54.7 | 40.5 | 43.7 | 29.6 | 28.1 | 18.5 | 14.5 | 9.3 | 46.7 | 34.8 |
| BiFDANet | 63.2 | 49.3 | 63.7 | 52.8 | 56.5 | 43.7 | 48.8 | 33.1 | 25.5 | 18.4 | 25.4 | 18.3 | 47.2 | 35.9 |
| IterDANet | 71.5 | 55.4 | 72.9 | 57.8 | 61.3 | 46.2 | 49.7 | 33.4 | 29.8 | 19.3 | 16.3 | 9.6 | 50.3 | 36.9 |
| FDANet | 69.6 | 54.9 | 72.4 | 57.3 | 56.3 | 43.2 | 48.4 | 31.9 | 27.3 | 18.6 | 19.5 | 10.8 | 48.9 | 36.1 |
| SAC | 69.7 | 54.3 | 72.8 | 57.5 | 60.5 | 45.8 | 49.3 | 33.2 | 30.2 | 19.5 | 18.2 | 10.1 | 50.1 | 36.7 |
| CPSL | 68.8 | 53.5 | 72.1 | 55.9 | 59.7 | 44.8 | 47.9 | 31.2 | 33.6 | 21.5 | 26.8 | 17.5 | 51.5 | 37.4 |
| ESPCN+AdvEnt | 67.8 | 54.5 | 68.8 | 54.9 | 54.1 | 38.0 | 46.3 | 31.2 | 34.2 | 22.4 | 27.6 | 16.1 | 49.8 | 36.2 |
| D-DBPN+AdvEnt | 68.4 | 54.6 | 69.5 | 55.2 | 54.8 | 38.2 | 46.7 | 31.6 | 35.4 | 24.5 | 28.4 | 16.4 | 50.5 | 36.8 |
| DASRSNet (ours) | **72.1** | **56.2** | **73.6** | **58.1** | 60.4 | 45.6 | 48.6 | 32.1 | **40.7** | **27.4** | **31.2** | **18.6** | **54.4** | **39.7** |

differences in various cities and spatial resolutions. We note that the spatial resolutions of HR POTSIRRG and POTSRGB images (5 cm) are about two times that of HR VAIH images (9 cm). Therefore, the spatial resolution of HR VAIH images is approximately two times that of LR POTSIRRG and POTSRGB images, which are downsampled from original HR images with a scaling factor of 4. Second, we learn to transfer the segmentation models from the HR VAIH dataset to LR POTSRGB dataset with medium domain shift. In this situation, the domain shift is caused by not only the differences in various cities and spatial resolutions but also the differences in the spectral bands. For instance, the POTSRGB dataset consists of blue, green, and red bands, while the VAIH dataset includes green, red, and infrared bands. Third, we probe the adaptability of our DASRSNet and the comparison models on the adaptation from the HR POTSRGB dataset to LR VAIH dataset with large domain shift, which ensues from the differences in various cities, spectral bands, and spatial resolutions. Furthermore, the spatial resolution shift becomes larger, i.e., the spatial resolution of HR POTSRGB images is near four times that of LR VAIH images. In this case, the small objects (e.g., cars) are even more difficult to be extracted and segmented by the classifier, which is transferred from the HR images. Then, to validate the effectiveness and robustness of our DASRSNet, we further train our algorithm on the same datasets with inherent spatial resolution differences, which avoids manual downsampling operations. We also attempt the proposed method on the real remote sensing images and give a discussion about these practical application scenarios. Besides the SS results, we also exhibit some SR results of our DASRSNet and compare SR methods. Finally, we study the advantages of our DASRSNet for the LR target images and small objects and discuss some failure cases.

*1) Small Domain Shift From HR VAIH to LR POTSIRRG:* First, we utilize the proposed DASRSNet and comparison models to eliminate the small domain gap between the HR VAIH dataset (source domain) and LR POTSIRRG dataset (target domain). Table II shows the quantitative segmentation results (F1 and IoU) for the proposed DASRSNet and comparing approaches.

From Table II, we can observe that the Deeplabv3+ (baseline) method performs worst among all the comparing approaches, whose mean F1 (mF1) and mean IoU (mIoU) are merely 26.2% and 15.7%, respectively. With the help of UDA techniques, the segmentation performances of other comparing approaches are enhanced apparently, which demonstrates the effectiveness and necessity of UDA techniques. We can find that the UDA models using self-supervised learning (e.g., IterDANet, SAC, and CPSL) have better performances than other comparing methods without SR, which shows the superiority of self-supervised learning. However, self-supervised learning cannot solve the spatial resolution shift problem well. In contrast, our DASRSNet reaches higher mF1 and mIoU values and outperforms the baseline model (Deeplabv3+) by 31.5% for mF1 and 26.2% for mIoU. Besides, our DASRSNet brings more performance improvements in all categories compared to other comparison models, and especially for category low vegetation, our DASRSNet obtains 50.0% F1 and 39.6% IoU performance improvement compared with the baseline model (Deeplabv3+). The above results highlight that the proposed DASRSNet is more conducive to reducing the domain gap caused by differences in spatial resolutions and various cities.

Besides, in Fig. 6, we draw the visual segmentation predictions for all the comparison approaches of Table II. From Fig. 6, one can observe that the Deeplabv3+ (baseline) method suffers from serious domain shift issues when transferring from the HR source domain to LR target domain. The predicted maps of Deeplabv3+ (baseline) are always noisy and unsatisfactory. With the aid of UDA, the visual predictions of other comparing methods are improved significantly. Among

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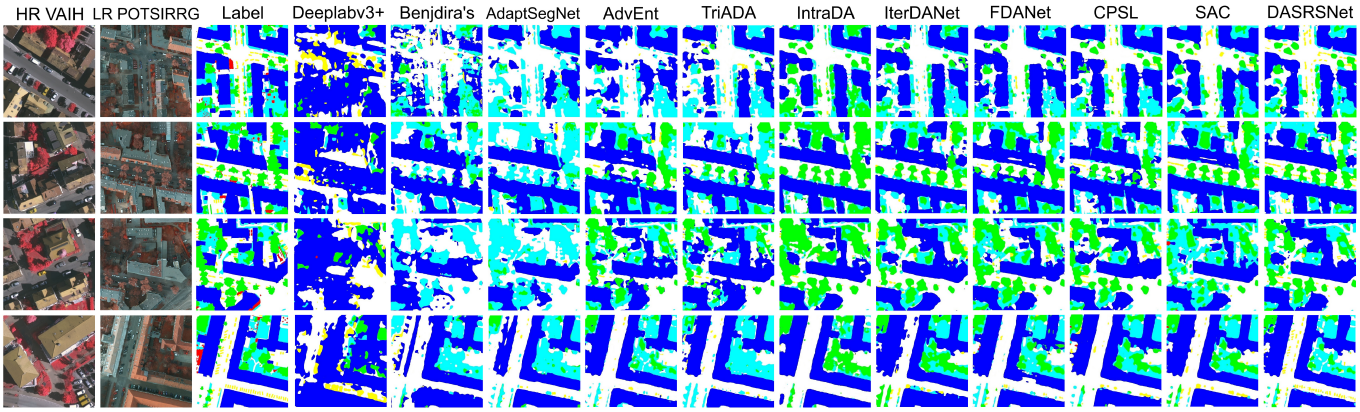


Fig. 6. Visualizations for segmentation predictions of all the comparing models in HR VAIH → LR POTSIRRG experiment. The first column is the samples for source HR VAIH images, while other columns are images, labels, and segmentation results of target LR POTSIRRG dataset.

TABLE III

SEGMENTATION RESULTS OF THE COMPARING SEGMENTATION MODELS ON THE ADAPTATION FROM THE HR VAIH DATASET TO LR POTSRGB DATASET. THE SCALING FACTOR BETWEEN SOURCE DOMAIN AND TARGET DOMAIN IS NEAR TWO TIMES. IMP SURF REPRESENTS IMPERVIOUS SURFACES AND LOW VEG REPRESENTS LOW VEGETATION

| Source: HR VAIH, Target: LR POTSRGB | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | Imp surf | | Building | | Low veg | | Tree | | Car | | Background | | Avg | |
| | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | mF1 | mIoU |
| Deeplabv3+(baseline) | 42.3 | 24.3 | 47.2 | 29.9 | 8.4 | 4.3 | 7.9 | 4.1 | 17.8 | 9.9 | 1.6 | 0.8 | 20.9 | 12.2 |
| Benjdira's | 65.1 | 52.8 | 63.7 | 51.9 | 48.1 | 34.1 | 45.8 | 28.7 | 18.3 | 10.1 | 12.9 | 6.9 | 42.3 | 30.8 |
| AdaptSegNet | 68.9 | 51.8 | 70.1 | 53.6 | 57.6 | 39.4 | 34.1 | 20.3 | 27.4 | 17.4 | 9.6 | 4.6 | 44.6 | 31.2 |
| AdvEnt | 70.3 | 53.9 | 71.8 | 56.5 | 53.8 | 36.8 | 40.2 | 26.4 | 25.5 | 16.7 | 10.2 | 5.3 | 45.3 | 32.6 |
| TriADA | 68.6 | 54.7 | 70.5 | 55.2 | **61.8** | **46.8** | **50.1** | **33.8** | 31.3 | 20.8 | 18.7 | 10.3 | 50.2 | 36.9 |
| IntraDA | 68.2 | 54.1 | 70.9 | 56.6 | 54.7 | 40.5 | 43.7 | 29.6 | 28.1 | 18.5 | 14.5 | 9.3 | 46.7 | 34.8 |
| BiFDANet | 63.2 | 49.3 | 63.7 | 52.8 | 56.5 | 43.7 | 48.8 | 33.1 | 25.5 | 18.4 | 25.4 | 18.3 | 47.2 | 35.9 |
| IterDANet | 71.5 | 55.4 | 72.9 | 57.8 | 61.3 | 46.2 | 49.7 | 33.4 | 29.8 | 19.3 | 16.3 | 9.6 | 50.3 | 36.9 |
| FDANet | 69.6 | 54.9 | 72.4 | 57.3 | 56.3 | 43.2 | 48.4 | 31.9 | 27.3 | 18.6 | 19.5 | 10.8 | 48.9 | 36.1 |
| SAC | 69.7 | 54.3 | 72.8 | 57.5 | 60.5 | 45.8 | 49.3 | 33.2 | 30.2 | 19.5 | 18.2 | 10.1 | 50.1 | 36.7 |
| CPSL | 68.8 | 53.5 | 72.1 | 55.9 | 59.7 | 44.8 | 47.9 | 31.2 | 33.6 | 21.5 | 26.8 | 17.5 | 51.5 | 37.4 |
| ESPCN+AdvEnt | 67.8 | 54.5 | 68.8 | 54.9 | 54.1 | 38.0 | 46.3 | 31.2 | 34.2 | 22.4 | 27.6 | 16.1 | 49.8 | 36.2 |
| D-DBPN+AdvEnt | 68.4 | 54.6 | 69.5 | 55.2 | 54.8 | 38.2 | 46.7 | 31.6 | 35.4 | 24.5 | 28.4 | 16.4 | 50.5 | 36.8 |
| DASRSNet (ours) | **72.1** | **56.2** | **73.6** | **58.1** | 60.4 | 45.6 | 48.6 | 32.1 | **40.7** | **27.4** | **31.2** | **18.6** | **54.4** | **39.7** |

all the UDA models, our DASRSNet acquires the best and most accurate segmentation results. This result verifies the superiority of the proposed DASRSNet over the SOTA UDA frameworks when facing spatial resolution shift problems.

*2) Medium Domain Shift From HR VAIH to LR POTSRGB:* Second, we attempt to mitigate the medium domain shift between the HR VAIH dataset (source domain) and the LR POTSRGB dataset (target domain) and report the quantitative segmentation results of the proposed DASRSNet and compare models in Table III.

Similarly, the Deeplabv3+ (baseline) method reaches worse performance against all the comparing UDA methods, with mIoU and mF1 as merely 12.2% and 20.9%, respectively. All the UDA methods obviously improve the segmentation performances of the baseline method without UDA, but the overall segmentation performances of most UDA methods slightly decrease compared with the first experiment. This is because the domain shift between the HR VAIH and LR POTSRGB datasets is larger than the domain shift between the HR VAIH and LR POTSIRRG datasets. Specifically, in this experiment, the features of the same class in the two datasets are quite different due to the differences in spectral bands. For example, low vegetations and trees are red in the HR VAIH dataset, while they are green in the LR POTSRGB dataset, as shown in Fig. 5. Also, such differences will significantly influence the features of the same categories, which greatly degrades the performance of most UDA models. Although the recently published UDA models (e.g., TriADA, IterDANet, and SAC) handle these issues and obtain the performance improvement on the categories of low vegetations and trees, our DASRSNet still performs favorably against all the comparison approaches in overall performance and brings over 2.3% mIoU and 2.9% mF1 improvement against these recently published SOTA models. Besides, our DASRSNet significantly outperforms other comparing models in category cars, which demonstrates that the proposed DASRSNet has more advantages in segmenting the small objects (e.g., cars) compared to the existing UDA methods.

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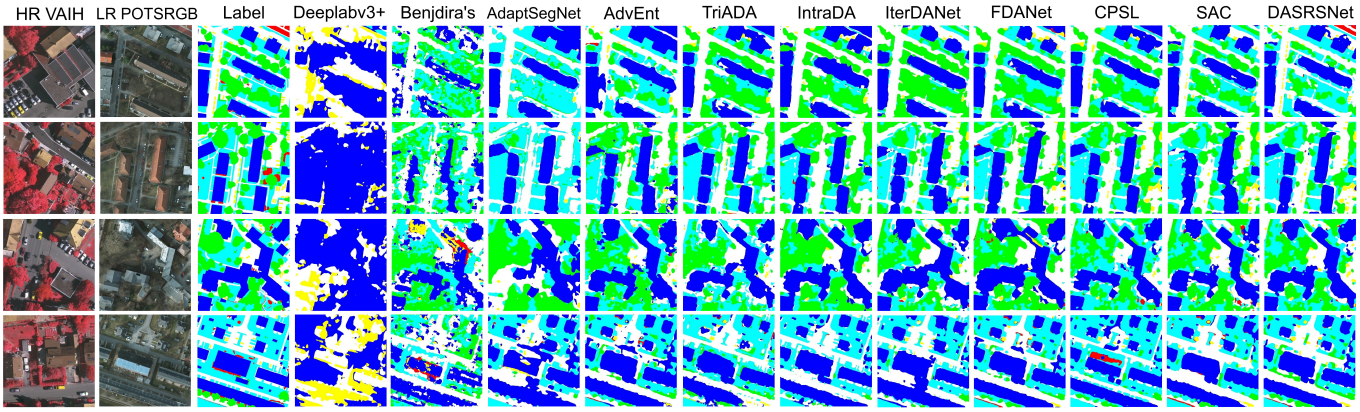


Fig. 7. Visualizations for segmentation predictions of all the comparing models in the HR VAIH → LR POTSRGB experiment. The first column is the samples for source HR VAIH images, while other columns are images, labels, and segmentation results of target LR POTSRGB dataset.

TABLE IV

SEGMENTATION RESULTS OF THE COMPARING SEGMENTATION MODELS ON THE ADAPTATION FROM THE HR POTSRGB DATASET TO LR VAIH DATASET. THE SCALING FACTOR BETWEEN SOURCE DOMAIN AND TARGET DOMAIN IS NEAR FOUR TIMES. IMP SURF REPRESENTS IMPERVIOUS SURFACES AND LOW VEG REPRESENTS LOW VEGETATION

| Source: HR POTSRGB, Target: LR VAIH | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | Imp surf | | Building | | Low veg | | Tree | | Car | | Background | | Avg | |
| | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | F1 | IoU | mF1 | mIoU |
| Deeplabv3+(baseline) | 37.9 | 23.4 | 36.1 | 22.0 | 8.1 | 4.6 | 6.9 | 3.5 | 0.4 | 0.2 | 2.0 | 1.0 | 15.2 | 9.1 |
| Benjdira's | 61.2 | 45.0 | 58.1 | 48.5 | 44.3 | 38.4 | 46.9 | 32.7 | 0.9 | 0.4 | 2.7 | 1.6 | 35.7 | 27.8 |
| AdaptSegNet | 66.4 | 50.8 | 60.3 | 49.1 | 41.4 | 35.5 | 46.5 | 32.6 | 0.5 | 0.3 | 2.2 | 1.7 | 36.2 | 28.3 |
| AdvEnt | 69.6 | 53.6 | 61.8 | 50.5 | 53.6 | 43.1 | 46.4 | 31.5 | 1.7 | 0.9 | 2.0 | 1.2 | 39.2 | 30.1 |
| TriADA | 71.5 | 54.7 | 70.9 | 56.6 | 51.7 | 42.6 | 48.7 | 32.8 | 1.8 | 0.8 | 3.2 | 2.4 | 41.3 | 31.7 |
| IntraDA | 70.2 | 54.1 | 67.3 | 53.7 | 44.9 | 38.7 | 42.1 | 30.4 | 0.9 | 0.3 | 8.3 | 5.1 | 38.9 | 30.4 |
| BiFDANet | 63.6 | 46.6 | 70.2 | 57.5 | 56.5 | 45.9 | 47.2 | 32.9 | 1.5 | 0.8 | 3.2 | 2.6 | 40.3 | 31.1 |
| IterDANet | 71.1 | 54.8 | 71.7 | 56.8 | 57.8 | 45.8 | 48.3 | 32.6 | 1.1 | 0.4 | 2.3 | 1.1 | 42.1 | 31.9 |
| FDANet | 69.3 | 53.9 | 72.2 | 57.5 | 51.4 | 42.4 | 46.6 | 31.5 | 1.7 | 0.8 | 3.1 | 2.3 | 40.7 | 31.4 |
| SAC | 70.8 | 54.5 | 71.4 | 56.3 | 56.2 | 45.2 | 46.2 | 31.3 | 1.3 | 0.7 | 4.6 | 1.8 | 41.8 | 31.6 |
| CPSL | 70.6 | 54.3 | 71.8 | 56.9 | 56.6 | 45.9 | 44.4 | 30.8 | 5.2 | 2.3 | 4.6 | 2.5 | 42.2 | 32.1 |
| ESPCN+AdvEnt | 67.7 | 53.3 | 68.9 | 54.8 | 54.7 | 42.5 | 44.0 | 28.0 | 15.7 | 6.6 | 4.4 | 2.8 | 42.6 | 31.3 |
| D-DBPN+AdvEnt | 68.1 | 53.7 | 69.5 | 55.4 | 54.9 | 42.9 | 44.6 | 28.6 | 17.1 | 8.2 | 6.3 | 4.2 | 43.4 | 32.2 |
| DASRSNet (ours) | **71.6** | **55.6** | **72.8** | **57.7** | **58.9** | **46.8** | **49.9** | **33.5** | **28.6** | **16.8** | **15.1** | **8.6** | **49.5** | **36.5** |

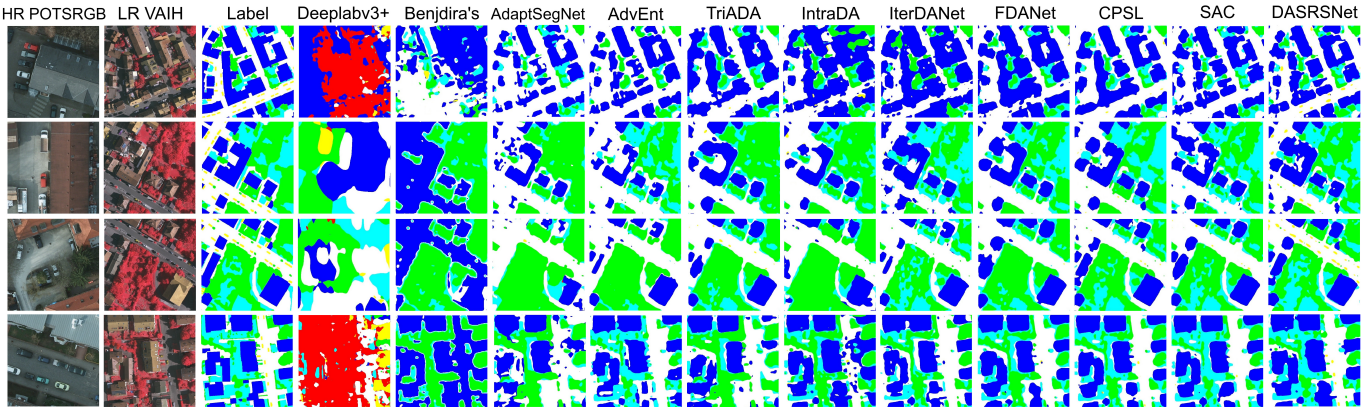


Fig. 8. Visualizations for segmentation predictions of all the comparing models in the HR POTSRGB → LR VAIH experiment. The first column is the samples for source HR POTSRGB images, while other columns are images, labels, and segmentation results of target LR VAIH dataset.

Fig. 7 shows the visual SS predictions of all the comparing frameworks of Table III. Similar to the above experiment, the Deeplabv3+ (baseline) outputs the most chaotic results without UDA for handling serious domain shift between the

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HR VAIH and LR POTSRGB datasets. With the help of UDA, the SS results become more accurate, and our DASRSNet can obtain the cleanest and most precise predictions for almost all categories, especially for small objects (e.g., cars), which highlights the superiority of our DASRSNet again.

*3) Large Domain Shift From HR POTSRGB to LR VAIH:* Third, to further confirm the robustness of the proposed DASRSNet in face of larger spatial resolution shift, we train to transfer SS models from the HR POTSRGB dataset (source domain) to LR VAIH dataset (target domain), which is more difficult to adapt than the previous two experiments. Because there are still differences in various cities and spectral bands, furthermore, the differences in spatial resolution become even larger (from scaling factor 2 to scaling factor 4), as shown in Fig. 5.

From Table IV, we can see that the mF1 and mIoU performances of other comparing UDA models drop severely compared with the above two adaptations, due to larger spatial resolution shift between source and target domains. Also, self-supervised learning used by recent SOTA UDA models (e.g., IterDANet, SAC, and CPSL) makes poor segmentation performance of small objects even worse. By contrast, the proposed DASRSNet still obtains superior performance against other UDA methods and reaches larger performance improvements, especially for small objects (e.g., car). Specifically, our DASRSNet shows an improvement of 4.3% in mIoU and 6.1% in mF1 over the second place. With respect to small objects (e.g., cars), the second best model does not improve the performance of category cars very much because of the huge spatial resolution shift between source and target domains. However, our DASRSNet apparently boosts the Deeplabv3+ (baseline) method by 16.6% in IoU and 28.2% in F1 on the category of cars, which is superior to recent SOTA UDA approaches.

As shown in Fig. 8, all the comparing frameworks suffer from huge spatial resolution shift problem between different domains and produce very poor predictions for the LR target images, especially for small objects (e.g., car). With the help of affinity of the SS and SR features, our DASRSNet can better extract the features for the LR target images and small objects within the LR target domain; thus, our framework can output more accurate and cleaner SS results compared to the existing UDA methods. In conclusion, the abovementioned experimental results demonstrate the superiority and robustness of our DASRSNet.

*4) Other UDA Scenarios:* In order to validate the effectiveness and robustness of the proposed DASRSNet when handling datasets with inherent spatial resolution differences, we conduct other two UDA experiments, i.e., original POTSIRRG to original VAIH and original POTSRGB to original VAIH, which avoid using manual downsampling operations. The differences between POTSIRRG and VAIH are various cities and spatial resolutions, while differences between POTSRGB and VAIH are various cities, spatial resolutions, and spectral bands. Table V reports the quantitative segmentation results of the proposed DASRSNet and all the comparing models. As shown in Table V, in both two UDA scenarios, the segmentation performances are apparently increased by UDA frameworks, which again indicates the effectiveness and necessity of UDA. Among all the UDA frameworks, our DASRSNet achieves the highest mF1 and mIoU performances and outperforms the second best method by 1.8% mIoU in the POTSIRRG → VAIH scenario and 0.7% mIoU in the POTSRGB → VAIH scenario. These experimental results verify the effectiveness and robustness of the proposed DASRSNet on the datasets with inherent spatial resolution differences.

TABLE V

SEGMENTATION RESULTS OF THE COMPARING SEGMENTATION MODELS ON THE ADAPTATION FROM THE ORIGINAL POTS DATASET TO ORIGINAL VAIH DATASET WITHOUT MANUAL DOWNSAMPLING OPERATIONS. THE SCALING FACTOR BETWEEN SOURCE DOMAIN AND TARGET DOMAIN IS NEAR TWO TIMES

| Method | POTSIRRG→VAIH | | POTSRGB→VAIH | |
|---|---|---|---|---|
| | mF1 | mIoU | mF1 | mIoU |
| Deeplabv3+ (baseline) | 28.6 | 16.5 | 22.5 | 13.9 |
| Benjdira's | 47.5 | 34.2 | 46.8 | 33.8 |
| AdaptSegNet | 49.8 | 36.4 | 48.4 | 35.1 |
| AdvEnt | 51.6 | 38.1 | 49.8 | 36.5 |
| TriADA | 54.1 | 39.8 | 51.9 | 38.2 |
| IntraDA | 54.4 | 40.3 | 52.8 | 38.7 |
| BiFDANet | 52.8 | 38.9 | 52.6 | 38.6 |
| IterDANet | 57.3 | 42.3 | 55.6 | 42.2 |
| FDANet | 54.9 | 40.4 | 53.7 | 39.5 |
| SAC | 55.9 | 41.7 | 54.5 | 40.1 |
| CPSL | 57.7 | 42.5 | 55.1 | 41.3 |
| ESPCN+AdvEnt | 55.3 | 41.2 | 54.3 | 39.8 |
| D-DBPN+AdvEnt | 56.2 | 42.1 | 54.8 | 40.6 |
| DASRSNet (ours) | **58.6** | **44.3** | **56.8** | **42.9** |

*5) Practical Application Scenarios:* In addition to the above benchmark datasets, we further utilize aerial images from real world to verify the generalization capacity of the proposed DASRSNet. The sampled Berlin and Zurich images are selected from the online Google Maps [51] with spatial resolutions of 9.1 and 10.1 cm, respectively. We train our DASRSNet and the comparing UDA methods to transfer the segmentation models from the HR POTSRGB dataset to Berlin and Zurich datasets via the same experimental settings as described earlier. As shown in Figs. 9 and 10, when adapting from the HR POTSRGB dataset to real-world LR aerial images, the proposed DASRSNet predicts more promising segmentation results against the baseline and recent SOTA UDA models.

*6) Results and Analysis for SR:* To better probe the SR ability of the DASR module, we draw a comparison between the proposed DASRSNet and several SOTA SR approaches: SRCNN [27], SRGAN [38], VDSR [41], FSRCNN [28], ESPCN [29], and D-DBPN [42]. Corresponding to the above HR VAIH → LR POTSIRRG, HR VAIH → LR POTSRGB, and HR POTSRGB → LR VAIH adaptation experiments, we train the comparing SR approaches on the source domain with the same training parameters as our DASRSNet. Then, we test the comparing SR approaches on the target images and report the quantitative results (PSNR and SSIM [49]) of

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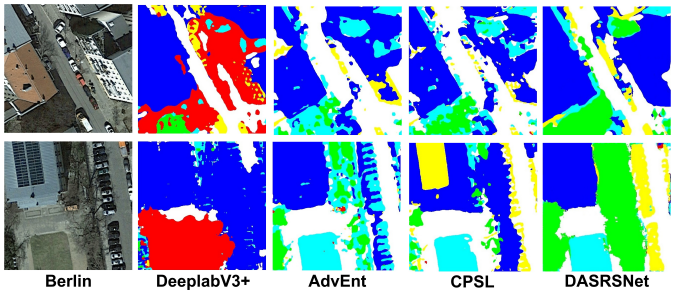


Fig. 9. Visual adaptive segmentation results of real aerial images (Berlin city) obtained from online Google Maps.

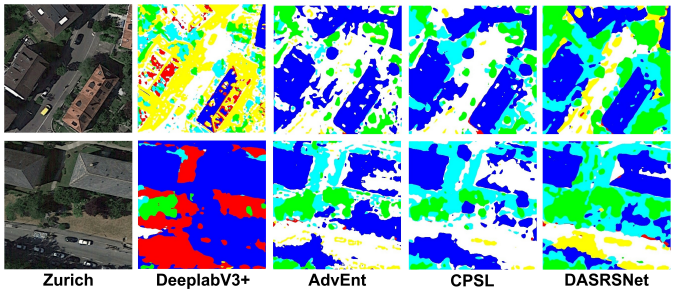


Fig. 10. Visual adaptive segmentation results of real aerial images (Zurich city) obtained from online Google Maps.

the above three experiments in Table VI. It is notable that the scaling factors for the three experiments are 2×, 2×, and 4×.

As shown in Table VI, our DASRSNet has better SR performance than other comparing SR approaches in most situations of all the three experiments. It is worth noting that ESPCN has the same decoder architecture as the proposed DASRSNet, but our DASRSNet reaches better performance than ESPCN and achieves 0.91-dB PSNR improvement over ESPCN in the HR POTSRGB → LR VAIH experiment. Moreover, D-DBPN achieves the comparable SR performance with our DASRSNet, at the cost of more complex network structure and more computational consumption. In comparison, our DASRSNet obtains superior SR results with less computational consumption and simpler network structure, which highlights the effectiveness and superiority of the proposed DASRSNet with shared encoder for SS and SR tasks.

Figs. 11–13 show the visual SR results corresponding to HR VAIH → LR POTSIRRG, HR VAIH → LR POTSRGB, and HR POTSRGB → LR VAIH experiments, respectively. One can observe that the great majority of comparing SR methods produce blurry reconstructed images, while the proposed DASRSNet can recover clearer and more accurate HR images among all the experiments, which highlights the effectiveness of DASR module of the proposed DASRSNet.

*7) Robustness to Small Object Segmentation:* When transferring from the HR source domain to LR target domain, the large spatial resolution shift makes it harder to segment the small objects (e.g., cars) than other objects, which drastically degenerates the performance of most commonly used UDA frameworks. The main reason is that the small objects (e.g., cars) in the LR target images take up much fewer pixels than the same objects in the HR source domain. Hence, when transferring from the HR source domain to LR target domain, it is more difficult for classifier trained on HR remote sensing

TABLE VI

SR RESULTS OF THE COMPARISON SR MODELS ON THE POTSIRRG, POTSRGB, AND VAIH DATASETS

| Method | VAIH→POTSIRRG | | VAIH→POTSRGB | | POTSRGB→VAIH | |
|---|---|---|---|---|---|---|
| | PSNR | SSIM | PSNR | SSIM | PSNR | SSIM |
| Bicubic | 30.09 | 0.892 | 29.75 | 0.884 | 25.63 | 0.791 |
| SRCNN | 32.87 | 0.908 | 32.16 | 0.897 | 26.92 | 0.804 |
| SRGAN | 33.46 | 0.916 | 31.83 | 0.893 | 27.67 | 0.812 |
| FSRCNN | 33.88 | 0.919 | 32.89 | 0.895 | 27.89 | 0.810 |
| VDSR | 34.48 | 0.927 | 33.51 | 0.911 | 28.54 | 0.821 |
| ESPCN | 33.72 | 0.921 | 32.73 | 0.908 | 27.81 | 0.818 |
| D-DBPN | 34.51 | 0.935 | 33.76 | **0.917** | 28.69 | 0.825 |
| DASRSNet | **34.57** | **0.939** | **33.78** | 0.916 | **28.72** | **0.827** |

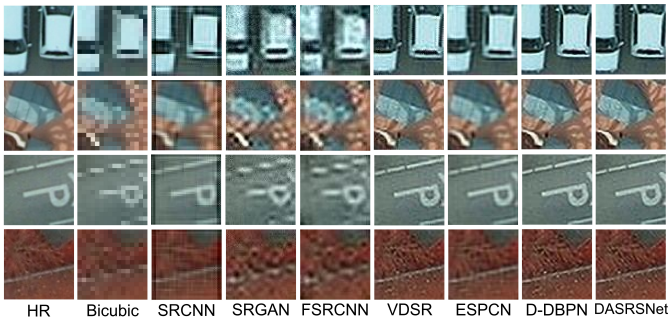


Fig. 11. Visual SR results for all the comparing models in the HR VAIH → LR POTSIRRG experiment.

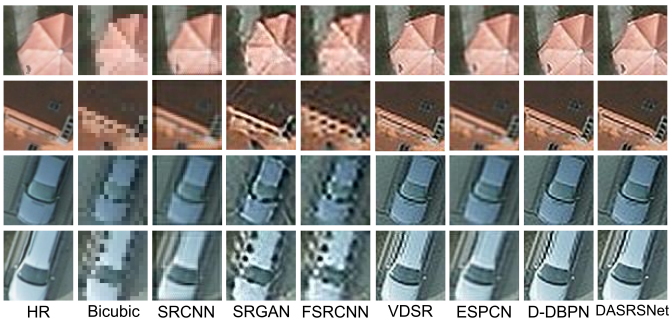


Fig. 12. Visual SR results for all the comparing models in the HR VAIH → LR POTSRGB experiment.

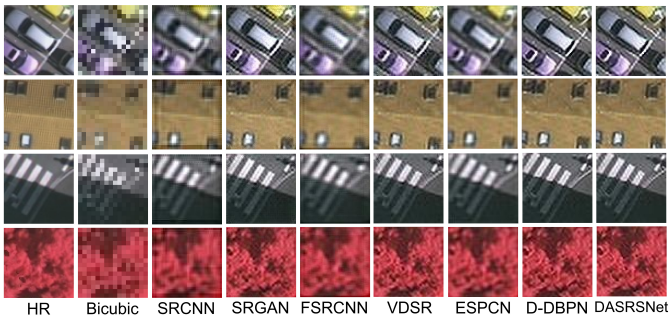


Fig. 13. Visual SR results for all the comparing models in the HR POTSRGB → LR VAIH experiment.

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images to extract textures and features from LR remote sensing images, which results in the failure of most recent UDA models to alleviate the domain shift for small objects.

As shown in Tables II and III, when the spatial resolution shift (about two times) between domains is not very large (such spatial resolution shift is usually ignored by existing UDA methods), TriADA wins the best performance on the category of cars among all the comparing models without SR, and it reaches an average 22.7% (between 24.5% and 20.8%) IoU and an average 35.3% (between 39.3% and 31.3%) F1 in the HR VAIH → LR POTSIRRG experiment and HR VAIH → LR POTSRGB experiment. However, in the HR POTSRGB → LR VAIH experiment, when the spatial resolution shift between the source and target domains becomes larger (from two times to four times), the comparing UDA approaches without SR obtain terrible performances on the category of cars, with IoU and F1 less than 2.3% and 5.2%, respectively, as shown in Table IV. This is indicative that the recent UDA models without SR are incapable of solving the serious spatial resolution shift problem for small objects. With the aid of SR, ESPCN + AdvEnt and D-DBPN + AdvEnt are not severely affected by the abovementioned issues and show apparent performance improvements on the category cars, due to the more HR and clearer target images recovered by SR methods. With respect to the proposed DASRSNet, the segmentation performance is further enhanced and our method shows a larger improvement of over 14.5% in IoU and 23.4% in F1 on the category cars compared to the UDA models without SR. This is because the proposed DASRSNet combines the SR features with HR SS features, which makes the SS features clearer for small objects.

Furthermore, in order to better compare our DASRSNet to other methods, we depict the close-up visual SR results and their corresponding SS results in Fig. 14. From Fig. 14, it can be seen that the comparing UDA model (AdvEnt) omits the small objects (cars) due to the blur and noise of the LR target remote sensing images. The ESPCN + AdvEnt method first reconstructs HR remote sensing images using SR and then conducts UDA between HR source images and reconstructed HR target images. With the aid of SR, this method can detect a part of cars effectively, but it still omits most cars, this is because the recovered HR target images are still slightly blurry and the reconstructed features are less relevant to the SS task. When applying more superior SR method (D-DBPN), the clearer recovered HR target images make D-DBPN + AdvEnt method segment more cars. However, most predictions are patchy, and the performance improvements are limited and less than our DASRSNet, as shown in Tables II–IV. One possible reason is that although the images recovered by superior D-DBPN are sharper and clearer in the SR task, not all of the HR details recovered by the D-DBPN are beneficial and helpful for SS task since the SS task and the SR task focus on different optimization objectives. In contrast, our DASRSNet successfully detects much more integral cars than other comparing methods and recovers comparable HR images for the SS task due to the sharing encoder for two different tasks and the affinity of SS and SR features in the proposed FA module, which demonstrates the effectiveness and superiority of our DASRSNet when handling the severe spatial resolution shift for small objects.

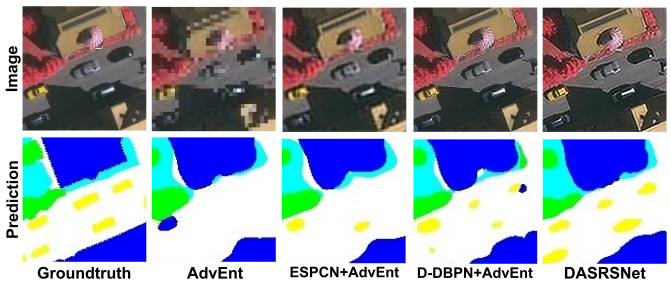


Fig. 14. Close-up visual adaptation result for small objects (e.g., cars) on the adaptation from the HR POTSRGB dataset to LR VAIH dataset. (From Left to Right) HR image and label, LR image and predicted result of AdvEnt, image recovered by ESPCN and predicted result of AdvEnt, image recovered by D-DBPN and predicted result of AdvEnt, and image recovered by DASRSNet and predicted result of DASRSNet (ours).

*8) Failure Case Discussion:* Although the proposed DASRSNet reaches the highest mIoU and mF1 values and obtains the most performance improvements for small objects (e.g., cars) in the above three experiments, it still exists a few defects and is poor at eliminating the domain shift caused by differences in spectral bands. As shown in Table III, when transferring from HR VAIH to LR POTSRGB dataset, BiFDANet and TriADA obtain higher IoU and F1 values than our method on the categories of trees and low vegetations. The primary reason is that the features of trees and low vegetations in the POTSRGB dataset differ a lot from the features of vegetations in the VAIH dataset. Specifically, the POTSRGB and VAIH images obtained from various sensors (domains) are composed of different types of spectral bands. In our experiments, VAIH images comprise green, red, and infrared bands, while POTSRGB images contain green, red, and blue bands. In this case, the low vegetations and trees are green in the POTSRGB dataset, while the vegetations turn red in the VAIH dataset due to the effect of the infrared band, which leads to serious differences in the features of the trees and low vegetations between POTSRGB and VAIH datasets. Consequently, the proposed DASRSNet is not good at extracting the domain-invariant features for the trees and low vegetations, which have quite different features in the POTSRGB and VAIH datasets. As a result, our DASRSNet obtains weak performances on the categories of trees and low vegetations, which exist severe domain shift resulting from differences in spectral bands.

### *D. Model Analysis*

In order to better explore the proposed DASRSNet, extensive ablation experiments are conducted on the VAIH, POTSIRRG, and POTSRGB datasets. First, the effectiveness and role of each component of the proposed DASRSNet are discussed. Second, we investigate the robustness of our DASRSNet with respect to various scaling factors between source and target domains. Third, we validate the convergence of our proposed network. Then, we study the effect of the values of hyperparameters in the DASRSNet. Finally, we exhibit

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TABLE VII

COMPONENT ANALYSIS OF OUR DASRSNET IN THE HR POTSRGB → LR VAIH EXPERIMENT. ADVENT DENOTES THE BASELINE UDA METHOD, WHILE DASS DENOTES THE PROPOSED DOMAIN ADAPTATION FOR SEMANTIC SEGMENTATION BRANCH. DASR DENOTES THE PROPOSED DOMAIN ADAPTATION FOR SUPER-RESOLUTION BRANCH AND FA DENOTES THE PROPOSED FEATURE AFFINITY MODULE. DASR-DASS DENOTES THE CONNECTING DASR MODULE AND DASS MODULE IN SERIES INSTEAD OF SHARING THE SAME ENCODER

| Source: HR POTSRGB, Target: LR VAIH | | | |
|---|---|---|---|
| Method | Components | mF1 | mIoU |
| Model-I | AdvEnt | 39.2 | 30.1 |
| Model-II | DASS | 41.8 | 31.4 |
| Model-III | DASR-DASS | 43.1 | 32.6 |
| Model-IV | DASS+DASR | 45.6 | 34.3 |
| Model-V | DASS+DASR+FA | **49.5** | **36.5** |

the computational consumption of our DASRSNet and other comparison models.

*1) Components Analysis:* As illustrated in Section III, the proposed DASRSNet has three key components: DASS branch, DASR branch, and FA module. In order to validate the role of the aforementioned components of the proposed DASRSNet, we repeatedly train the ablation models with different components on the adaptation from the HR POTSRGB dataset (source domain) to LR VAIH dataset (target domain) and present the quantitative results of different ablation models in Table VII. For the baseline model Model-I, we adopt the popular UDA model AdvEnt [35], which has the similar architecture to our DASS branch. Model-II denotes merely utilizing the DASS branch of our DASRSNet. Model-III represents using the DASR branch before the DASS branch so as to restore HR target images for the DASS branch, and these two branches are connected in series rather than sharing the encoder. With respect to Model-IV, the DASS branch and DASR branch are combined by sharing the same encoder, as shown in Fig. 2. Finally, Model-V corresponds to the proposed DASRSNet.

As can be observed in Table VII, Model-II obtains higher mIoU and mF1 performances than Model-I without the help of class balance loss [43], which shows the effectiveness of our DASS branch. Then, Model-III, which performs SR before the DASS branch, leads to further performance improvement compared with Model-II. Such results indicate that SR is beneficial to the SS task when transferring segmentation models from HR images to LR remote sensing images. Furthermore, Model-IV increases the mIoU from 32.6% of the ablation Model-III to 34.3%, which demonstrates that the architecture combined the DASS branch and DASR branch by sharing the same encoder is superior to the architecture where these two branches are connected in series. Finally, Model-V adopting FA module reaches the best performance among all the ablation models, which highlights the necessity of the FA module.

Moreover, we also show the visual segmentation features of various ablation models in Fig. 15. Specifically, we utilize the popular visualization tool (matplotlib library in python),

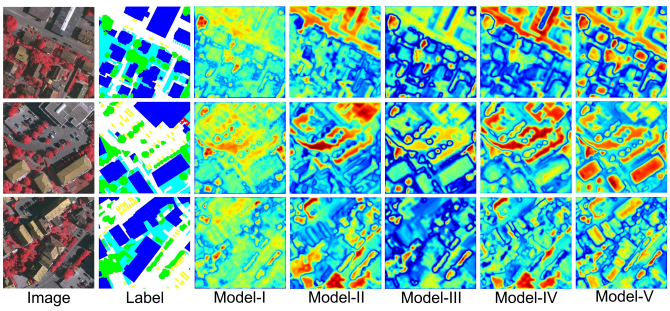


Fig. 15. Visual feature-level segmentation predictions of all the ablation models in Table VII in the HR POTSRGB → LR VAIH experiment.

which colorizes the segmentation features with various colors according to their numerical values (the closer the values, the more similar the colors), to draw the visual features obtained from the last convolutional layer (before softmax function) of the segmentation decoder $S$. From Fig. 15, it can be seen that the segmentation features of Model-I (baseline) are the most noisy and worst among all the ablation models, which means that the existing UDA methods are poor at dealing with the spatial resolution shift between the HR source domain and the LR target domain. With the addition of different components of the proposed DASRSNet, the segmentation features of ablation models gradually become cleaner and better. Also, for all the ablation models, our DASRSNet (Model-V) obtains the cleanest and best segmentation features, which shows the effectiveness and superiority of the proposed framework again.

*2) Robustness to Different Scaling Factor:* The scaling factor between the HR source domain and LR target domain is an important element for domain shift caused by a spatial resolution shift problem. The larger scaling factor between the source domain and target domain, the huger domain shift between these two domains. To assess the robustness of the proposed DASRSNet in the face of different scaling factors between source and target domains, we conduct several experiments on the adaptation from the HR VAIH dataset to LR POTSIRRG dataset with various scaling factors, including 2, 4, 6, and 8. In particular, we maintain the original spatial resolution of the source HR VAIH images while downsampling the POTSIRRG images through bicubic interpolation with degradation factors 2, 4, 6, and 8. Table VIII shows the quantitative segmentation results of our DASRSNet and the popular UDA model AdvEnt.

As shown in Table VIII, when the scaling factor between source and target domains increases from 2 to 8, the segmentation performance of the comparison model AdvEnt decreases sharply from 51.8% mF1 and 37.3% mIoU to 37.4% mF1 and 28.7% mIoU. Moreover, for category cars, the IoU performance of AdvEnt method severely drops to 1.3% when the scaling factor reaches 8. This is mainly because a large number of textures and features will be lost when the spatial resolution of remote sensing image becomes lower. Also, the larger the degradation factor is, the more textures and features are lost, especially for small objects (e.g., cars). In contrast, when the scaling factor becomes larger, the segmentation performance of the proposed DASRSNet only degenerates slightly, and

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TABLE VIII

INFLUENCE OF SCALING FACTOR BETWEEN HR SOURCE DOMAIN AND LR TARGET DOMAIN ON UDA METHODS. ADVENT DENOTES THE BASELINE UDA METHOD WITHOUT SR

| Source: HR VAIH, Target: LR POTSIRRG | | | | | |
|---|---|---|---|---|---|
| Scaling Factor | Method | $F1_{car}$ | $IoU_{car}$ | mF1 | mIoU |
| 2 | AdvEnt | 34.6 | 21.5 | 51.8 | 37.3 |
| | DASRSNet | 46.8 | 32.4 | 60.4 | 42.6 |
| 4 | AdvEnt | 26.4 | 16.8 | 48.1 | 35.2 |
| | DASRSNet | 42.3 | 28.7 | 57.7 | 41.9 |
| 6 | AdvEnt | 15.7 | 9.6 | 43.2 | 32.6 |
| | DASRSNet | 35.9 | 23.8 | 53.8 | 39.7 |
| 8 | AdvEnt | 2.6 | 1.3 | 37.4 | 28.7 |
| | DASRSNet | 28.1 | 16.4 | 48.6 | 36.2 |

our DASRSNet brings larger performance improvements compared with the baseline UDA method AdvEnt, especially for category cars. When the scaling factor changes from 2 to 8, the performance improvement between our DASRSNet and the baseline UDA method AdvEnt improves from 8.6% mF1 and 5.3% mIoU to 11.2% mF1 and 7.5% mIoU, and for category cars, the performance improvement increases significantly from 12.2% F1 and 10.9% IoU to 25.5% F1 and 15.1% IoU. These indicate that benefiting from the affinity of SS features and SR features, our DASRSNet is more robust and conducive to handling the larger scaling factor between the source and target domains, especially for small objects (e.g., cars).

*3) Convergence Analysis:* In this section, to confirm the convergence of the proposed DASRSNet, which performs multitask adversarial training not only in the output space but also in the image space, we depict the training loss curves of our DASRSNet on the adaptation from the HR POTSRGB dataset to LR VAIH dataset in Fig. 16. Fig. 16 shows the training curves of the total training loss $\mathcal{L}_{DASRSNet}$ and each component loss (i.e., segmentation loss $\mathcal{L}_{\text{seg}}$, adversarial loss for segmentation $\mathcal{L}_{\text{adv}}^{\text{seg}}$, FA loss $\mathcal{L}_{\text{fa}}$, SR loss $\mathcal{L}_{\text{sr}}$, and adversarial loss for SR $\mathcal{L}_{\text{adv}}^{sr}$) against the number of epochs. From Fig. 16, we can find that for the DASS branch, the training losses (segmentation loss $\mathcal{L}_{\text{seg}}$ and adversarial loss $\mathcal{L}_{\text{adv}}^{\text{seg}}$) decrease gradually and achieve convergence as the training epoch increases. With respect to the DASR branch, the training losses (SR loss $\mathcal{L}_{\text{sr}}$ and adversarial loss $\mathcal{L}_{\text{adv}}^{sr}$) show a similar trend and achieve a stable state. In general, during the training phase, the total training loss $\mathcal{L}_{\text{DASRSNet}}$ first drops sharply and then slowly converges with the increasing epochs. It is indicative that the adversarial training of the DASS branch (output space) and DASR branch (image space) does not interfere with each other, and the proposed DASRSNet can reach convergence in the case of multitask adversarial training.

*4) Hyperparameter Analysis:* In this section, we illustrate the settings of the hyperparameters ($\lambda_{\text{adv}}^{\text{seg}}$, $\lambda_{\text{sr}}$, $\lambda_{\text{adv}}^{sr}$, and $\lambda_{\text{fa}}$) in the proposed DASRSNet and study the effects of the values of these parameters, as shown in Table IX.

First, we train our network with only segmentation loss $\mathcal{L}_{\text{seg}}$ as the baseline. Second, we add the adversarial loss for segmentation $\mathcal{L}_{\text{adv}}^{\text{seg}}$ to perform UDA training between the target and source domains. As shown in Table IX, we can see that our model achieves the best performance when we set $\lambda_{\text{adv}}^{\text{seg}} = 0.01$, and a smaller or larger value of parameter $\lambda_{\text{adv}}^{\text{seg}}$ will degrade the segmentation performance. Therefore, we set $\lambda_{\text{adv}}^{\text{seg}} = 0.01$ in the following experiments. Third, with respect to adding the SR loss $\mathcal{L}_{\text{sr}}$, our model obtains the largest performance improvement with $\lambda_{\text{sr}} = 0.1$, if the weight of $\lambda_{\text{sr}}$ becomes too lower or too higher, the tradeoff between the segmentation task and SR task will be affected, which will influence the segmentation performance. Hence, we set $\lambda_{\text{sr}} = 0.1$. Then, when considering the influence of $\lambda_{\text{adv}}^{sr}$ for the adversarial loss for SR $\mathcal{L}_{\text{adv}}^{sr}$, we find that our model performs best when $\lambda_{\text{adv}}^{sr} = 0.001$. The smaller or larger weight of $\lambda_{\text{adv}}^{sr}$

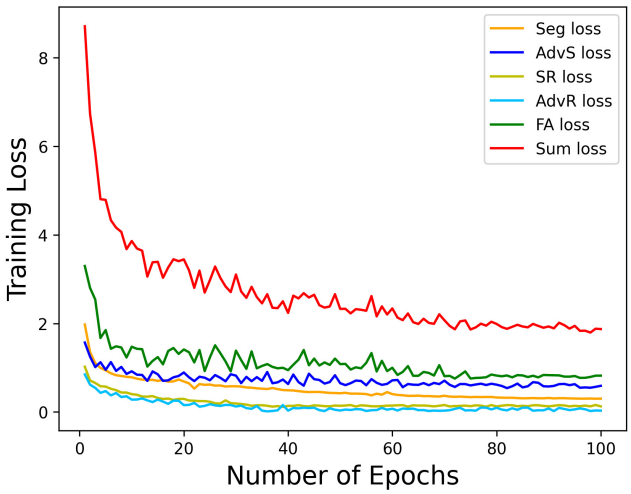


Fig. 16. Training loss curves of the total loss and all the component losses. Seg loss, AdvS loss, FA loss, SR loss, AdvR loss, and Sum loss represent the segmentation loss $\mathcal{L}_{\text{seg}}$, adversarial loss for segmentation $\mathcal{L}_{\text{adv}}^{\text{seg}}$, FA loss $\mathcal{L}_{\text{fa}}$, SR loss $\mathcal{L}_{\text{sr}}$, adversarial loss for SR $\mathcal{L}_{\text{adv}}^{sr}$, and total training loss $\mathcal{L}_{\text{DASRSNet}}$, respectively.

TABLE IX

HYPERPARAMETER ANALYSIS OF DASRSNET ON THE ADAPTATION FROM THE HR VAIH DATASET TO LR POTSRGB DATASET

| Source: HR VAIH, Target: LR POTSRGB | | | | | | |
|---|---|---|---|---|---|---|
| Loss | $\lambda_{adv}^{seg}$ | $\lambda_{sr}$ | $\lambda_{adv}^{sr}$ | $\lambda_{fa}$ | mF1 | mIoU |
| $\mathcal{L}_{seg}$ | 0 | 0 | 0 | 0 | 22.7 | 13.4 |
| $\mathcal{L}_{adv}^{seg}$ | 0.005 | 0 | 0 | 0 | 44.8 | 33.1 |
| | 0.01 | 0 | 0 | 0 | **46.2** | **33.5** |
| | 0.02 | 0 | 0 | 0 | 45.7 | 33.2 |
| | 0.05 | 0 | 0 | 0 | 44.3 | 32.7 |
| $\mathcal{L}_{sr}$ | 0.01 | 0.05 | 1 | 0 | 46.1 | 34.1 |
| | 0.01 | 0.1 | 5 | 0 | **47.4** | **34.8** |
| | 0.01 | 0.2 | 5 | 0 | 47.1 | 34.6 |
| | 0.01 | 0.5 | 10 | 0 | 45.8 | 33.7 |
| $\mathcal{L}_{adv}^{sr}$ | 0.01 | 0.1 | 0.0005 | 0 | 49.5 | 36.7 |
| | 0.01 | 0.1 | 0.001 | 0 | **50.3** | **37.2** |
| | 0.01 | 0.1 | 0.002 | 0 | 49.8 | 37.0 |
| | 0.01 | 0.1 | 0.005 | 0 | 48.9 | 36.3 |
| $\mathcal{L}_{fa}$ | 0.01 | 0.1 | 0.001 | 0.5 | 52.6 | 39.0 |
| | 0.01 | 0.1 | 0.001 | 1 | **54.4** | **39.7** |
| | 0.01 | 0.1 | 0.001 | 2 | 53.7 | 39.3 |
| | 0.01 | 0.1 | 0.001 | 5 | 51.8 | 38.6 |

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TABLE X
MODEL PERFORMANCE (MIOU), MODEL PARAMETERS (PARAMS), TRAINING TIME, AND TEST TIME FOR EACH PATCH OF COMPARING FRAMEWORKS ON THE HR POTSRGB AND LR VAIH DATASETS

| Source: HR POTSRGB, Target: LR VAIH | | | | |
|---|---|---|---|---|
| Method | mIoU (%) | Params (MB) | Training time (h) | Test time (s) |
| Benjdira's | 27.8 | 578.0 | 8.94 | 0.26 |
| AdaptSegNet | 28.3 | 220.5 | 4.10 | 0.26 |
| AdvEnt | 30.1 | 220.5 | 4.12 | 0.26 |
| TriADA | 31.7 | 229.7 | 6.43 | 0.26 |
| IntraDA | 30.4 | 232.0 | 9.05 | 0.26 |
| BiFDANet | 31.1 | 787.0 | 11.27 | 0.62 |
| IterDANet | 31.9 | 798.5 | 15.38 | 0.26 |
| FDANet | 31.4 | 429.5 | 8.61 | 0.26 |
| SAC | 31.6 | 418 | 8.28 | 0.26 |
| CPSL | 32.1 | 418.4 | 8.36 | 0.26 |
| ESPCN+AdvEnt | 31.3 | 220.7 | 5.64 | 0.37 |
| D-DBPN+AdvEnt | 32.2 | 260.2 | 7.23 | 0.51 |
| DASRSNet | 36.5 | 232.2 | 5.28 | 0.26 |

will cripple the performance of the classifier. Thus, we set $\lambda_{\text{adv}}^{sr} = 0.001$ for the following experiments. Finally, for $\lambda_{\text{fa}}$, the DASRSNet has the best performance with $\lambda_{\text{fa}} = 1$. If $\lambda_{\text{fa}}$ is too small or large, the final performance of segmentation model will be degraded. As a result, we set $\lambda_{\text{adv}}^{\text{seg}} = 0.01$, $\lambda_{\text{sr}} = 0.1$, $\lambda_{\text{adv}}^{sr} = 0.001$, and $\lambda_{\text{fa}} = 1$ as the final values of parameters adopted in the experiments.

*5) Computational Analysis:* In this section, we comprehensively compare the proposed DASRSNet with existing UDA methods in terms of computational consumption and segmentation performance. We train all the comparison frameworks and our DASRSNet on the single NVIDIA Tesla P40 GPU using the same architectures and training parameters. Table X reports the test time (per patch), training time, model parameters, and mIoU performance in the HR POTSRGB→LR VAIH adaptation experiment.

From Table X, we can see that when segmenting the same image patch during the testing phase, most frameworks cost about the same test time due to the same structure of segmentation model used in the experiments. It is notable that our DASRSNet will not consume any additional time at the test stage because our DASRSNet can remove the DASR branch (SR decoder) when merely conducting a segmentation task. With respect to the training stage, the image translation-based methods (Benjdira's, BiFDANet, and IterDANet) cost more training time and model parameters, due to the usage of extra image translation network. Except for these three methods, IntraDA, SAC, CPSL, and FDANet require more training time than other frameworks listed in Table X. This is because SAC, IntraDA, and CPSL adopt a two-stage training mode, which executes further training for the target classifier based on the existing one-stage UDA training. For FDANet, it utilizes much more loss functions and aligns more spaces (i.e., image, feature, and output spaces) than most UDA methods. Therefore, their segmentation performances can be enhanced against the one-stage UDA methods (e.g., AdaptSegNet and AdvEnt) at the cost of training time and model parameters. In comparison, the proposed DASRSNet takes less training time and model parameters but achieves better segmentation performance, which benefits from the shared structure for the DASS branch and DASR branch. Although the proposed DASRSNet requires more training time and model parameters than some one-stage UDA methods (e.g., AdaptSegNet, AdvEnt, and TriADA), our DASRSNet brings much more performance improvement without additional test time. In conclusion, our DASRSNet can reach a better balance between the computational cost and model performance than the recent UDA frameworks.

## V. CONCLUSION

In this article, we present a novel two-stream UDA network called DASRSNet for SR-aided segmentation of remote sensing images so as to further solve the serious spatial resolution shift problem and severe domain shift for small objects in remote sensing images. The proposed DASRSNet is composed of three modules: DASS branch, DASR branch, and FA module. Specifically, the DASS module extracts the domain-invariant SS features of different domains and aligns the domain shift in the output level, while the DASR module learns the domain-invariant SR features and eliminates the domain gap in the image space. Furthermore, to boost the affinity of SS features and SR features on both source and target domains, we propose a novel FA loss in the FA module, which is designed to strengthen the correlation between the DASS branch and the DASR branch. The extensive experiments on the benchmark datasets, which consist of aerial images obtained from different sensors and cities, indicate that the proposed DASRSNet can reach better segmentation performance over the existing UDA frameworks when facing the spatial resolution shift problem, especially for small objects. Moreover, we conduct numerous ablation experiments to better explore the SS ability of our DASRSNet. In our future work, the proposed DASRSNet will be further optimized with more powerful techniques, such as weakly supervised domain adaptation [8] and adaptive segmentation with multiple fields of view [52], so as to deal with the domain shift resulted from various spectral bands and more complex inherent spatial resolution shift problems in practical applications (e.g., transferring from aerial images to satellite images).

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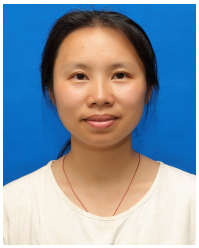

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